Name.....

Combination.....

P525/1
Chemistry
Paper 1
2¾ hours

TARGET COMMUNITY COLLEGE SCHOOL

Uganda Advanced Certificate of Education

S5 END OF TERM 1 EXAMINATIONS 2022
CHEMISTRY
PAPER 1
TIME: 2 hours 45 minutes

INSTRUCTIONS TO CANDIDATES;

- Answer **all** questions in section A and **six** questions in section B
- All questions must be answered in the spaces provided
- The periodic table, with relative atomic masses is attached at the end of the paper.
- Molar gas constant, R = $8.314JK^{-1}mol^{-1}$
- Molar volume of a gas at s.t.p is 22.4 litres

FOR THE EXAMINER'S USE ONLY																	Total
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	

SECTION A: (46 MARKS)

Answer **all** the questions in this section.

1. (a) Explain what is meant by the term radioactive decay. (1 mark)

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(b) Complete the following nuclear equations (3 marks)

(i) $^{232}_{90}Th \rightarrow \ ^{4}_{2}He \ +$……

(ii) $^{42}_{19}K + ^{4}_{2}He \rightarrow ^{1}_{0}n +$ ……

(iii) $^{243}_{97}Bk + ^{1}_{0}n \rightarrow ^{1}_{1}H \ +$ ……

(c) (i) State the decay law (1 mark)

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(ii) A sample of fresh wood gave 20 counts $g^{-1}s^{-1}$ due to the disintegration of carbon-14. A sample of wood from a prehistoric site in East Africa gave 5 counts $g^{-1}s^{-1}$. If the half-life of carbon-14 is 5700 years, what is the age of the wood? (3 marks)

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2. (a) Define the term isotopes (1 mark)

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(b) Copper has a relative atomic mass of 63.55 and consists of two isotopes of mass 63 and 65. Calculate the percentage abundance of the isotopes (3 marks)

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3. A gaseous hydrocarbon Q contains 90% carbon. The density of Q is 1.785×10^{-3} gcm^{-3} at s.t.p. Determine:

(i) the empirical formula of Q. (2½ marks)

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(ii) the molecular formula of Q (3 marks)

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4. (a) Write the electronic configurations of the following; (4 marks)

(i) Nitrogen atom

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(ii) Oxide ion

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(iii) Chlorine atom

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(iv) Potassium ion

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5. (a) Define the term oxidation state (1 mark)

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(b) Determine the oxidation state of;

(i) Zn in ZnO_2^{2-} (1 mark)

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(ii) Al in AlO_2^- (1 mark)

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(iii) Br in BrO_3^- (1 mark)

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(iv) Cu in $CuCl_2^-$ (1 mark)

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6. (a) What is meant by the term atomic radius? (1 mark)

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(b). Explain how each of the following factors affect atomic radius

(i) Nuclear charge (2 marks)

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(ii) Screening effect (2 marks)

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(c). The ionic radius of Fe^{2+} is bigger than that of Fe^{3+}. Explain this observation

(2 marks)

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7. (a) What is meant by the term first electron affinity? (1 mark)

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(b) Write an equation to represent the first electron affinity of Oxygen (1 mark)

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(c) Explain why the value for the second electron affinity of oxygen is positive while the value for the first electron affinity is negative (2 marks)

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8. (a) State Graham's law of gaseous diffusion (1 mark)

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(b) Two pieces of glass wool, one soaked in an amine and the other soaked in concentrated hydrochloric acid were placed at opposite ends of a glass tube 1m long. After some time, a white ring was observed 0.52m from the end containing the amine. Determine the molecular mass of the amine. (3 marks)

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9. (a) Define a functional group (1 mark)

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(b). Name the following functional groups (4 marks)

(i) $-NH_2$

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(ii) C = C

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(iii) $-NO_2$

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(iv) –COOH

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SECTION B: (54 MARKS)

Attempt only six (6) questions from this section.

10. Define each of the following types of structural isomerism giving an example in each case.

(a) Chain Isomerism (3 marks)

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(b) Position Isomerism (3 marks)

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(c) Functional group Isomerism (3 marks)

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11.(a) Define the term critical temperature. (1 mark)

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(b) The diagram in **Figure 1** shows isothermals of a gas.

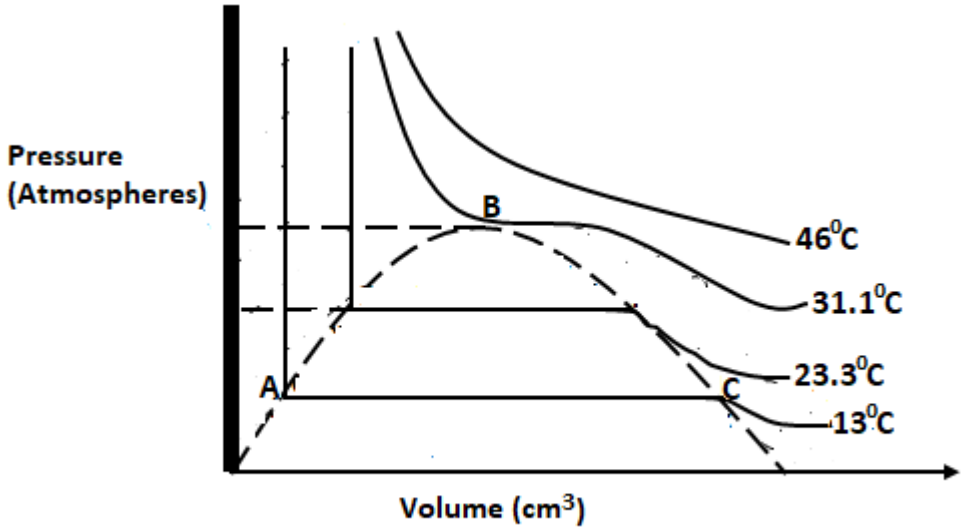

Figure 1

(i) On the Figure 1, indicate the following phases of the gas.

- Liquid (½ marks)
- Vapour (½ marks)

(ii) What is the critical temperature of the gas? (½ marks)

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(iii) Which isothermal is in accordance with Boyle's law? (½ marks)

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(c) State;

(i) what the region ABC represents (½ marks)

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(ii) The conditions necessary for liquefaction of a gas. (1 mark)

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(d)(i) Define Partial Pressure (1 mark)

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(ii) State Dalton's law of Partial Pressure (1 mark)

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(iii) A mixture consists of 2.2g of carbon dioxide and 3.5 g of nitrogen gas. If the total pressure is 10 atmospheres, calculate the partial pressure of each of the components (C = 12, O = 16, N = 14) (2½ marks)

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12. (a) Explain what is meant by each of the following terms.

(i) Order of a reaction (1 mark)

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(ii) Molecularity of a reaction (1 mark)

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(b) The table below shows kinetic data for the following reaction;

$$X + Y \longrightarrow Z$$

[X] ($moldm^{-3}$)	[Y] ($moldm^{-3}$)	Initial rate ($moldm^{-3}s^{-1}$)
0.100	0.50	2.0×10^{-3}
0.100	0.25	2.0×10^{-3}
0.050	0.25	1.0×10^{-3}
0.025	0.25	5.0×10^{-4}

State the order of reaction with respect to each of the reactant species X and Y. In each case give a reason for your answer.

(i) X; Order; ..... (½ mark)

Reason; (1 mark)

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(ii) Y; Order; ..... (½ mark)

Reason; (1 mark)

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(iii) Write the rate equation for the reaction above. State what each symbol means in the written expression. (2½ marks)

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(iv) Calculate the rate constant for the reaction and indicate the units. (1½ marks)

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13. (a) Define the term ionisation energy (1 mark)

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(b) Write an equation to show the first ionisation energy of magnesium. (1 mark)

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(c) The table below shows the ionization energies (in $kJmol^{-1}$) of five elements lettered A, B, C, D, and E.

Elements	1st I.E	2nd I.E	3rd I.E	4th I.E
A	500	4600	6900	9500
B	746	1500	7700	10500
C	630	1600	3000	4800
D	900	1800	14800	21000
E	580	1800	2700	11600

(i) Which one of the above elements is most likely to form an ion with a charge of $^{+}1$. Give a reason for your answer. (2 marks)

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(ii) State two elements which belong to the same group in the periodic table. (1 mark)

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(iii). Identify the group to which the elements you have stated in c(ii) belong. (1 mark)

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(iv). Write the formula of the chloride of element E. (1 mark)

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(d) Briefly explain how nuclear charge affects the ionisation energy (2 marks)

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14. (a) 1.363g of compound Y containing carbon, hydrogen and bromine on complete combustion gave 1.10g of carbon dioxide and 0.45g of water. When 0.35g of Y was vaporized, it occupied 39.5 cm^3 at 20^0C and 750 mmHg.

(i) Calculate the simplest formula of Y (1½ marks)

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(ii) Calculate the molecular formula of Y (2½ marks)

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(b) When 142cm^3 of a hydrocarbon Q of molecular mass 58g was exploded with excess oxygen and cooled to room temperature, the volume of the residual gas was 694cm^3. On treatment with concentrated potassium hydroxide solution, the volume decreased to 126cm^3.

(i) Determine the molecular formula of Q. (3½ marks)

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(ii) Write the names and structural formula of all the isomers of Q (1½ marks)

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15. (a) Define the following;

(i) Reducing agent (1 mark)

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(ii) Oxidizing agent (1 mark)

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(b) Give two examples of;

(i) Reducing agents (1 mark)

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(ii) Oxidizing agents (1 mark)

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(b) A solution was made by dissolving 5.3g of anhydrous sodium carbonate to make $250cm^3$ of solution. $25cm^3$ of this solution required $12.5cm^3$ of hydrochloric acid solution. Determine the molarity of the hydrochloric acid solution. (5 marks)

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THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Tc 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 Tl 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															
			139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 69	173 Yb 70	175 Lu 71
			227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

THE END